Maker Utopias

Speculative design futuring in a polycrisis through miniatures

Cindy Kohtala, Umeå Institute of Design < cindy.kohtala@umu.se >
Rickard Åström, Umeå Institute of Design, Interaction Lab
Sara Rylander, Department of Creative Studies, Teacher Education, Sloydlab
Magnus Wink, Department of Creative Studies, Teacher Education, Sloydlab

Introduction: why makers?

Maker Utopias is an artistic research project exploring open design and open hardware technology subcultures and their discourses and practices oriented to 'sustainability'. Makers and hackers experiment with open source solutions and ways of organizing that offer alternatives to global mass production and consumption. These communities act translocally (Loorbach et al., 2020) and encourage others to actively participate in locally relevant and socially good design and production. Their narratives envision replacing empty consumerism with more empowering, creative, circular, democratic and community-based production in shared workshops (fab labs and makerspaces), according to local needs, materials and resources.

Yet realizing such ideals as being more 'local', 'sustainable' and 'democratizing' is not always straightforward, as our longitudinal ethnographic research has shown (Kohtala, 2017; 2018). Maker culture is global with regional flavours (Braybrooke & Jordan, 2017; Kohtala et al., 2020); even if relatively niche and nascent, it is regarded as a social movement and as a compelling research topic in several fields, such as how it has impacted the design profession and design education (e.g. Bianchini & Maffei, 2020) and how it serves as a material indicator of citizen participation in society (Rosa et al., 2017; Smith et al., 2017; Papadopoulos, 2018). In our research endeavours, we are interested in understanding and articulating the effects of maker culture in industrial transition, as a technology- and product-oriented social movement (Hess, 2005; 2016).

We create provocative utopian dioramas of shared community technology workshops, to examine DIY maker and hacker cultures in new ways. We ask, how can we use critical design techniques to analyse phenomena and relations with technology in materially embodied and engaged ways? What does it look like and what does it take, to build sustainable solar societies – as "Real Utopias" (Olin Wright, 2013) – in the interstices of unsustainable macro structures? Perhaps we as a society can only create such utopias in miniature form.

Methods: why miniatures?

Artists, designers, architects and researchers use arts-based and design methods to make tiny utopias: making alternative futures tangible and possible by imagining them and bringing them (or parts of them) into being, from science fiction stories to games, fictional tableaux, installations, interactive devices, architectural models, and so on (Bell & Pahl, 2017; Browning, 2022; Kohtala, 2024). Design researchers have tackled ethical and philosophical issues related to our troubled relations to technology in hands-on experiments with diverse groups, as "critical making" (Matto, 2011). They problematize technomyths, encourage design and material literacies, and critically discuss ecological issues through "high-low tech" making (Kadish & Dulic, 2015).

We position ourselves within Design Research, using hands-on critical making as a method informed by Science & Technology Studies (STS) frameworks to guide our explorations and analysis of our work, which we document through shared discussions and autoethnographic notes. We borrow from "prototyping the past" research methods in design and HCI (Sayers, 2015) where aesthetic and conceptual ideas about the past, present and future of technology and local production are enacted in small scale, imitating natural history museum dioramas. We also use digital fabrication techniques, echoing the means that makers use. In this study, we have created a diorama of a 'green fab lab' exhibited with two photographs, one of an existing green fab lab in Barcelona and another a detailed macro view of the diorama, in a peer-reviewed exhibition in UmArts (Kohtala et al., 2025).

The work is inspired by studies of miniatures in art history and utopian studies (Davy & Dixon, 2019; Fiduccia, 2022), as well as the illustrations of utopian community spaces (such as collective gardens and community workshops) by anarchist illustrator Clifford Harper (Boyle & Harper, 1976). Miniatures allow us to play with the histories of making, such as the radical design movements active during the energy crisis of the 1970s, its presents, and its speculative futures.

The project is ongoing and is a collaboration between the Umeå Institute of Design (UID) and the Department of Creative Studies, Teacher Education, and their workshops, UID's Interaction Lab and Creative Studies' Sloydlab. This study received seed funding from UmArts as a Small Visionary Project in 2024-25 (Kohtala et al., 2025).

Below: *Untitled*, model, C. Kohtala, R. Åström, S. Rylander, M. Wink, 2025. Foam, MDF, plywood, cardstock, paper, miniature tools, miniature scale, dollhouse clock, paint.

Bottom left: *Untitled*, photograph, C. Kohtala. Valldaura Self-Sufficiency Lab, Barcelona, Spain, 30 August 2016.

Bottom right: *Untitled*, photograph, C. Kohtala. Green Fab Lab model, detail, 9 June 2025.







Conclusions: why utopias?

Maker culture is often seen as techno-utopian, and even the solarpunk visions and experiments by ecologically oriented maker communities are seen as too little in the face of climate crisis, hypercapitalism and the sixth mass extinction – a true polycrisis (Kohtala et al., 2020; Kohtala, 2024). We critically examine the phenomenon of making and maker culture as representing the participatory turn: an emerging mode of localized, place-based, participatory production.

Miniatures and dioramas create fictional worlds that startle with their realism in relation to their scale. They easily combine material techniques, conventional craft, digital fabrication and electronics circuitry, as well as reclaimed objects such as toys. When photographed they convey an other-worldly aesthetic that is appropriate to exploring utopias and anti-utopias.

The project has also allowed us to re-examine our fieldwork in makerspaces, fab labs, DIYbio labs and sustainable materialist activist communities, providing insight, for example, into the articulation work (Star & Strauss, 1999; Kohtala, 2018) needed to build community, design open-design and open-hardware community-based products that can be fabricated locally, such as 3D printers, open-source beehives, or wikihouses, and share these designs. It also serves as a novel communication method for illustrating research findings, using the charm and allure of miniatures to entice, surprise and inform. Our future studies and exhibitions aim to stimulate dialogue among design and craft communities about the ethics of digital-physical making and the Real Utopias of local production.

References

Bell, D. M., & Pahl, K. (2017). Co-production: towards a utopian approach. *International Journal of Social Research Methodology*, 21(1), 105–117. https://doi.org/10.1080/13645579.2017.1348581

Bianchini, M., & Maffei, S. (2020). Facing the Fourth Industrial Revolution: Empowering (human) design agency and capabilities through experimental learning. *Strategic Design Research Journal*, 13(1), 72–91. https://doi.org/10.4013/sdrj.2020.131.06

Boyle, G., & Harper, P. (Eds.). (1976). *Radical Technology*. Wildwood House.

Braybrooke, K., & Jordan, T. (2017). Genealogy, Culture and Technomyth: Decolonizing Western Information Technologies, from Open Source to the Maker Movement. *Digital Culture & Society*, 3(1), 25–45. https://doi.org/10.14361/dcs-2017-0103

Browning, B. (2022). *The Miniaturists*. Duke University Press.

Davy, J., & Dixon, C. (eds.). (2019). *Worlds in Miniature: Contemplating Miniaturisation in Global Material Culture*. UCL Press.

Fiduccia, J. (2022). Scale of the Nation: Alberto Giacometti's Miniature Monument. *Art History*, 45(1), 126–156. https://doi.org/10.1111/1467-8365.12633

Hess, D. J. (2005). Technology- and Product-Oriented Movements: Approximating Social Movement Studies and Science and Technology Studies. *Science, Technology, & Human Values*, 30(4), 515–535. https://doi.org/10.1177/0162243905276499

Hess, D.J. (2016). *Undone Science: Social Movements, Mobilized Publics, and Industrial Transitions*. MIT Press.

Kadish, D., & Dulic, A. (2015). Crafting Sustainability: Approaching Wicked Environmental Problems through High–Low Tech Practice. *Digital Creativity*, 26(1), 65–81. https://doi.org/10.1080/14626268.2014.998682

Kohtala, C. (2017). Making "Making" Critical: How Sustainability is Constituted in Fab Lab Ideology. *The Design Journal*, 20(3), 375–394. https://doi.org/10.1080/14606925.2016.1261504

Kohtala, C. (2018). The Sociomateriality of FabLabs: Configurations of a Printing Service or Counter-Context? *Journal of Peer Production*, 12. http://peerproduction.net/issues/issue-12-makerspaces-and-institutions/peer-reviewed-papers/the-sociomateriality-of-fablabs/

Kohtala, C. (2024). Solarpunk as a Maker Imaginary. In P. Troxler & C. Kohtala (eds.), *Proceedings of the Fab 24 Research Papers Stream: Fabricating Equity*. Rotterdam University of Applied Sciences. https://doi.org/10.5281/zenodo.13221344

Kohtala, C., Boeva, Y., & Troxler, P. (2020). Introduction: Alternative Histories in DIY Cultures and Maker Utopias. *Digital Culture & Society*, 6, 5–34. https://doi.org/10.14361/dcs-2020-0102

Kohtala, C., Åström, R., Rylander, S., Wink, M. (2025). Maker Utopias (installation). UmArts Summer Exhibition 2025, Umeå University (June 11-18). https://urn.kb.se/resolve?urn=urn:nbn:se:umu:diva-243618

Levitas, R. (2013). *Utopia as Method*. Palgrave Macmillan.

Loorbach, D., Wittmayer, J., Avelino, F., von Wirth, T., & Frantzeskaki, N. (2020). Transformative innovation and translocal diffusion. *Environmental Innovation and Societal Transitions*, 35, 251–260. https://doi.org/10.1016/j.eist.2020.01.009

Papadopoulos, D. (2018). *Experimental practice: Technoscience, alterontologies, and more-than-social movements*. Duke University Press.

Ratto, M. (2011). Critical Making: Conceptual and Material Studies in Technology and Social Life. *The Information Society*, 27(4), 252–260. https://doi.org/10.1080/01972243.2011.583819

Rosa, P., Ferretti, F., Guimarães Pereira, Â., Panella, F., & Wanner, M. (2017). Overview of the Maker Movement in the European Union (JRC Technical Reports No. EUR 28686 EN). Publications Office of the European Union.

Sayers, J. (2015). Prototyping the Past. *Visible Language*, 49.

Smith, A., Fressoli, M., Abrol, D., Arond, E., & Ely, A. (2017). *Grassroots Innovation Movements*. Routledge.

Star, S. L., & Strauss, A. (1999). Layers of Silence, Arenas of Voice: The Ecology of Visible and Invisible Work. *Computer Supported Cooperative Work (CSCW)*, 8(1–2), 9–30.

Wright, E. O. (2013). Transforming Capitalism through Real Utopias. *American Sociological Review*, 78(1), 1–25. https://doi.org/10.1177/0003122412468882